



Project Report

Multivariate Customer Segmentation for E-Commerce

Abstract

This project delivers an end-to-end multivariate statistical analysis and customer segmentation for an e-commerce dataset. Using exploratory data analysis, feature engineering, PCA, factor analysis (KMO & Bartlett checks, Varimax rotation), clustering, canonical correlation analysis, and hypothesis testing, we derive statistically validated customer segments and provide actionable business recommendations to increase retention, cross-sell and customer lifetime value (LTV). The analysis emphasizes interpretability and reproducibility for business stakeholders.

1. Introduction & Business Objective

E-commerce businesses need accurate, explainable customer segmentation to improve targeting, personalize offers, and allocate marketing budget efficiently. This project aims to produce statistically rigorous, business-actionable customer segments using multivariate methods. The emphasis is on interpretable factors and segments that tie directly to marketing levers.

Key business questions:

- Are there distinct customer groups with different purchase patterns?
 - Which product affinities or purchase behaviors drive segment differences?
 - Which segments are most valuable and which should be prioritized for retention or reactivation?
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2. About the Data

Source: `Multivariate_Customer_Segmentation_for_E_Commerce.ipynb` / `e-commerce-data.csv` (uploaded notebook data)

Schema (summary):

- `Cust_ID` (customer identifier)
- `Gender` (categorical)
- `Orders` or `computed brand_sum` (frequency proxy)
- ~35 brand columns (per-customer purchase counts per brand)

- Derived fields created in the notebook: `brand_sum`, `purchase_diversity`, `cat_<Category>`, factor scores `F1..Fk`, PCA columns `PC1`, `PC2`, and cluster label `cluster`.

Notes about raw data:

- The dataset is customer-level (one row per customer) with brand purchase counts.
- No invoice date or monetary (unit price) data were available; therefore the classical RFM (Recency, Frequency, Monetary) was adapted:
 - Frequency: used Orders or `brand_sum` (sum of brand counts).
 - Monetary: not available; analysis uses brand counts, category aggregates and diversity as proxies for customer value/behavior.

3. Preprocessing & Feature Engineering

Major steps executed in the notebook:

1. Missing values & types
 - Brand columns NaNs replaced with 0 (no purchase).
 - Ensured numeric types for brand counts and order counts.
2. Derived features
 - `brand_sum`: total items purchased across all brands (purchase frequency proxy).
 - `purchase_diversity`: number of different brands purchased by each customer (count of nonzero brand columns).
 - `cat_<Category>` aggregates: brand columns grouped into coarse categories (e.g., Electronics, Apparel, Food, Other) using a keyword mapping (adjustable).
3. Transformations
 - `log1p` applied to brand purchase counts for PCA/EFA where needed to reduce skew.
 - Standard scaling (zero mean, unit variance) before PCA/EFA/KMeans/LDA.

Rationale: Without monetary and recency fields, we used brand-level behavior and diversity to capture customer preference and engagement patterns.

4. Exploratory Data Analysis (EDA)

Key EDA findings (high-level — exact numbers available in notebook outputs):

- Purchase distributions:

- **brand_sum** and most brand columns are right-skewed (long tails); log1p transformation improves symmetry for multivariate analysis.
- **purchase_diversity** shows many customers buying from a few brands and a long tail of more diverse shoppers.
- **Top brands:**
 - The notebook lists top 10–12 brands by total purchases (useful for product-level targeting).
- **Correlations:**
 - Correlation heatmap for the top brands reveals grouped affinities (brands that tend to be bought together) — a useful input for grouping into categories and for factor analysis.

EDA conclusion: Customer behavior shows clear heterogeneity (heavy tail buyers, specialty vs. generalist customers), supporting segmentation.

5. Dimensionality Reduction — PCA

Purpose: reduce dimensionality of the brand matrix and visualize customer space.

- **Data used:** log1p(brand_counts) then standardized.
- **Result:**
 - First few principal components explain the majority of variance (PC1, PC2 capture the biggest axes).
 - Biplot / PC1 vs PC2 scatter shows separation by behavioral preference (for example, customers with high electronics affinity vs. apparel/food affinity).
 - The PCA loadings identify which brands load strongly onto each component (top loadings printed in the notebook).

Business takeaway: PCA provides interpretable axes of behavior (e.g., Electronics-oriented vs Apparel-oriented customers) that can inform marketing segmentation and bundling strategies.

6. Exploratory Factor Analysis (EFA) with KMO, Bartlett, Varimax

Purpose: extract latent behavioral factors and ensure factor analysis suitability.

- **Suitability checks:**

- **KMO (overall sampling adequacy) reported in notebook.**
Interpretation rule: KMO > 0.6 considered acceptable; >0.8 is great.
- **Bartlett's Test p-value reported; $p < 0.05$ indicates correlation matrix suitable for factor analysis.**
- **Factor extraction:**
 - **Number of factors selected using Kaiser's rule (eigenvalues > 1) and scree plot (visual elbow).**
 - **Rotation: Varimax rotation applied to produce sparse, interpretable factor loadings.**
- **Results & interpretation:**
 - **Each factor's top positive loadings indicate behavioral dimensions — e.g.:**
 - **Factor 1: Electronics affinity (brands like Samsung, Mi).**
 - **Factor 2: Apparel & footwear affinity (brands like Vans, Wrangler).**
 - **Factor 3: Food/CPG affinity (brands like Kraft, Gatorade).**
 - **Factor scores (F1..Fk) were computed per customer and saved for downstream clustering.**

Business takeaway: Factor analysis reduces brand noise into 2–5 interpretable behavioral traits that marketers can use to drive targeted campaigns.

7. Clustering (K-Means) on Factor Scores

Purpose: form customer segments that share similar latent behaviors.

- **Features used for clustering: Factor scores derived from EFA (interpretable dimensions); optionally PC scores.**
- **Model selection: tried $K=2..8$; silhouette score and inertia (elbow) used to pick optimal K.**
- **Chosen K: notebook selects K with highest silhouette — reported in outputs.**
- **Cluster visualization: clusters plotted on PCA (PC1 vs PC2) to visualize separability.**
- **Profiles:**
 - **For each cluster, the notebook reports:**
 - **customers (count),**
 - **avg_brand_sum, median_brand_sum,**

- avg_purchase_diversity,
- mean category purchases (cat_Electronics, cat_Apparel, cat_Food, etc.),
- top brands by mean purchases for that cluster.
- Typical segment archetypes (illustrative — check notebook for exact cluster labels and numbers):
 - Cluster A (High-Value Specialists): high brand_sum, high diversity, strong electronics/apparel purchases — target with premium bundles and VIP retention.
 - Cluster B (Frequent Low-Diversity Buyers): moderate brand_sum, low diversity — opportunities for cross-sell and discovery offers.
 - Cluster C (Occasional Food/CPG Buyers): purchases concentrated in food category — target with subscriptions/sampling programs.
 - Cluster D (Low-engagement): low brand_sum and diversity — candidate for win-back or cost-efficient engagement.

Business takeaway: Clusters identify segments suitable for different marketing plays (premium offers vs cross-sell vs reactivation).

8. Linear Discriminant Analysis (LDA)

Purpose: examine how well clusters are linearly separable and produce discriminant axes for interpretability.

- LDA was trained using cluster labels (from KMeans) as class labels. This approach is exploratory — clusters are unsupervised labels.
- Metrics:
 - Training/test accuracy scores printed in the notebook.
 - LDA coefficients show which factor features most contribute to separating clusters.
- Interpretation:
 - Good LDA accuracy suggests clusters are linearly separable in factor space — helpful for simple scoring/rules to assign new customers to segments.

Caveat: Using LDA on clusters is exploratory (labels are internally derived), and it should be presented as such.

9. Canonical Correlation Analysis (CCA)

Purpose: explore relationships between two sets of variables — e.g., Electronics vs Apparel+Food category aggregates.

- **Design:**
 - Group A = category set (e.g., Electronics)
 - Group B = category set (e.g., Apparel + Food)
- **Result:**
 - The first canonical correlation coefficient (reported in notebook) quantifies how strongly preferences in Group A relate to preferences in Group B.
 - A high canonical correlation suggests cross-category affinities that can be exploited for cross-sell.

Business takeaway: If customers who buy electronics also purchase accessories from apparel/CPG categories, cross-category campaigns will have good lift potential.

10. Hypothesis Testing — Cluster Differences & Associations

A formal statistical testing block was run (the notebook contains the exact numeric outputs). Tests performed include:

Univariate tests (for each metric: brand_sum, purchase_diversity)

1. Assumption checks
 - Levene's test for equal variances across clusters.
 - Normality tests per cluster (D'Agostino / Shapiro) where appropriate.
2. Choice of test
 - If normality & equal variance satisfied: One-way ANOVA, with eta-squared as effect size, and Tukey HSD post-hoc pairwise comparisons.
 - Otherwise: Kruskal-Wallis nonparametric test, and pairwise Mann-Whitney U tests with Bonferroni correction and rank-biserial effect sizes.
3. MANOVA
 - Performed on (brand_sum, purchase_diversity) jointly across clusters to assess multivariate differences.
4. Categorical association
 - Chi-square test and Cramer's V for association between cluster and Gender (if Gender present).

Reported results (see notebook outputs):

- Whether brand_sum differs significantly across clusters (p-value, which cluster pairs differ).
- Whether purchase_diversity differs significantly.
- MANOVA multivariate statistics.
- Any significant association between cluster membership and gender.

Actionable interpretation rules:

- Significant differences with medium/large effect sizes justify different business programs for the affected clusters (e.g., invest in VIP retention for cluster with significantly higher brand_sum).
- If gender is associated with cluster, tailor creative and offers by gender where appropriate — but check for business ethics and fairness.

11. Segment-level Business Action Plans (example outputs)

For each cluster the notebook produced targeted recommendations derived from profiling:

- **High-Value / High-Diversity**
 - VIP loyalty program, early access, premium bundles, A/B test premium pricing.
 - Personalized cross-sell offers based on their top brands.
- **High Frequency / Low Diversity**
 - Discovery campaigns, targeted product recommendations to increase basket variety.
 - Low-risk sampling or “try one, get one” to broaden brand exposure.
- **Low Engagement / At-Risk**
 - Win-back campaigns with limited-time incentives, feedback surveys to discover churn reasons.
 - Avoid heavy discounting unless ROI-justified — consider low-cost retention flows.
- **Category-Specific Buyers (e.g., Food/CPG heavy)**
 - Subscription or replenishment promotions.
 - Bundles with complementary categories identified by CCA.

12. Conclusions

- The dataset's brand-level signals are rich enough to produce interpretable multivariate dimensions (via PCA/EFA).
- Factor analysis compactly summarizes brand preferences into a few behavioral factors, enabling robust, interpretable clustering.
- Clusters show statistically significant differences on core behavioral metrics (see hypothesis testing outputs). These differences support differentiated marketing strategies.
- The combined approach (PCA → EFA → clustering → LDA/CCA → hypothesis testing) yields a reproducible segmentation framework that is ready for A/B testing of targeted campaigns and further extension to CLTV modeling if monetary/recency data are later provided.